The Living World

Exam Context: Comprehensive study guide and assessment aligned strictly to the NEET UG syllabus. Covers perspectives on life and biodiversity, binomial nomenclature, taxonomy versus systematics, and the complete taxonomic hierarchy with biological examples.

1. Learning Objectives & Chapter Overview

After studying this tutorial, you will be able to:

- Understand the historical shift from an anthropocentric view of biology to systematic descriptions and biodiversity conservation.
- Recognize the contributions of **Ernst Mayr** to evolutionary biology and the biological species concept.
- Distinguish between the technical and philosophical inquiries into the nature of life.
- Apply the universal rules of **Binomial Nomenclature** established by Carolus Linnaeus and governed by codes such as ICBN and ICZN.
- Differentiate clearly between **identification**, **nomenclature**, **classification**, **taxonomy**, and **systematics**.
- Arrange and analyze the seven obligate **taxonomic categories** (from Species to Kingdom) and understand how shared characteristics vary across the hierarchy.
- Recall the exact taxonomic categories of key representative organisms: Man, Housefly, Mango, Wheat, Potato, Brinjal, Makoi, Petunia, Datura, Lion, Tiger, Leopard, Cat, and Dog.

2. Perspectives on the Living World & Pioneer Profile

2.1 Nature of Biology and Human Perception

Biology is the science of life forms and living processes. The living world comprises an amazing diversity of organisms inhabiting extraordinary environments—including cold mountains, deciduous forests, oceans, freshwater lakes, deserts, and hot springs. Whether observing a galloping horse, migrating birds, a valley of flowers, an attacking shark, ecological conflict and cooperation among populations of a community, or the molecular traffic inside a cell, the living world evokes a deep sense of wonder.

- **Early Human Perception:** Early man could easily perceive the difference between inanimate matter and living organisms. He deified some inanimate matter (wind, sea, fire, etc.) and some animals and plants. The common feature of all such animate and inanimate objects was the sense of *awe or fear* that they evoked.
- **Anthropocentric View:** The formal description of living organisms, including human beings, began much later in human history. Societies that indulged in an *anthropocentric* (human-centered) view of biology registered limited progress in biological knowledge.
- **Spin-off of Systematic Studies:** Monumental and systematic descriptions of life forms necessitated detailed systems of identification, nomenclature, and classification. The biggest spin-off of such studies was the recognition of the sharing of similarities among living organisms both **horizontally** (among existing contemporaries) and **vertically** (across evolutionary generations).
- **Conservation Impact:** The revelation that all present-day living organisms are related to each other and to all organisms that ever lived on earth humbled humankind and led to cultural movements for the conservation of biodiversity.

2.2 Pioneer Profile: Ernst Mayr (1904–2004)

Ernst Mayr (born 5 July 1904 in Kempten, Germany; died in 2004 at the age of 100) was a renowned Harvard University evolutionary biologist regarded as one of the 100 greatest scientists of all time.

- **Epithet:** Called *"The Darwin of the 20th century"*.

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- **Academic Tenure:** Joined Harvard's Faculty of Arts and Sciences in 1953 and retired in 1975, assuming the title *Alexander Agassiz Professor of Zoology Emeritus*.
- **Research Scope:** Over his nearly 80-year career, his work spanned ornithology, taxonomy, zoogeography, evolution, systematics, and the history and philosophy of biology.
- **Key Scientific Contributions:**
  1. He almost single-handedly made the **origin of species diversity** the central question of evolutionary biology.
  2. He pioneered the currently accepted definition of a **biological species**.
- **The "Triple Crown" of Biology:** Mayr was awarded the three prizes widely regarded as the triple crown of biology:
  - **Balzan Prize** (1983)
  - **International Prize for Biology** (1994)
  - **Crafoord Prize** (1999)

2.3 The Question: "What is Living?"

When reflecting on what life is, the question contains two implicit questions within it:

1. **Technical Question:** Seeks the answer to what living is as opposed to the non-living (identifying defining characteristics of living organisms).
2. **Philosophical Question:** Seeks the answer to what the purpose of life is. As scientists, biologists do not attempt to answer this philosophical question; instead, scientific inquiry focuses on the technical question of what constitutes "living".

3. Diversity in the Living World

3.1 Biodiversity

Around us exists a vast variety of potted plants, insects, birds, pets, and microscopic organisms invisible to the naked eye. If you increase the observation area (for instance, visiting a dense forest), the range and variety of observable organisms increase substantially. Each different kind of plant, animal, or organism that you see represents a **species**.

Key Fact for NEET: The number of species that are currently known and described ranges between 1.7 million **and** 1.8 million (or 17–18 lakhs). This refers to **biodiversity**—the number and types of organisms present on earth. As we explore new areas and even old ones, new organisms are continuously being identified.

3.2 Need for Nomenclature and Identification

Organisms are known in different regions by their local (vernacular) names, which vary from place to place even within a single country. Without a standardized system, communicating about specific organisms across regions would create severe confusion.

- **Nomenclature:** The process of standardizing the naming of living organisms such that a particular organism is known by the same name all over the world.
- **Identification:** Nomenclature is only possible when an organism is described correctly and we know to what organism the name is attached. This process is called identification.

To ensure that each organism has only one scientific name globally—and that such a name has not been used for any other known organism—international codes have been established:

- **ICBN:** *International Code for Botanical Nomenclature* (provides agreed principles and criteria for naming plants; Note: the preposition is specifically "for").
- **ICZN:** *International Code of Zoological Nomenclature* (evolved by animal taxonomists for naming animals; Note: the preposition is specifically "of").

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3.3 Binomial Nomenclature

Given by **Carolus Linnaeus** and practiced by biologists worldwide, **Binomial Nomenclature** is the two-word naming system that provides each known organism with a scientific name consisting of two components:

$$\text{Scientific Name} = \text{Generic name} + \text{specific epithet}$$

For example, the scientific name of mango is *Mangifera indica*. Here, *Mangifera* represents the **genus** (Generic name) while *indica* represents a particular species, or **specific epithet**.

Universal Rules of Binomial Nomenclature

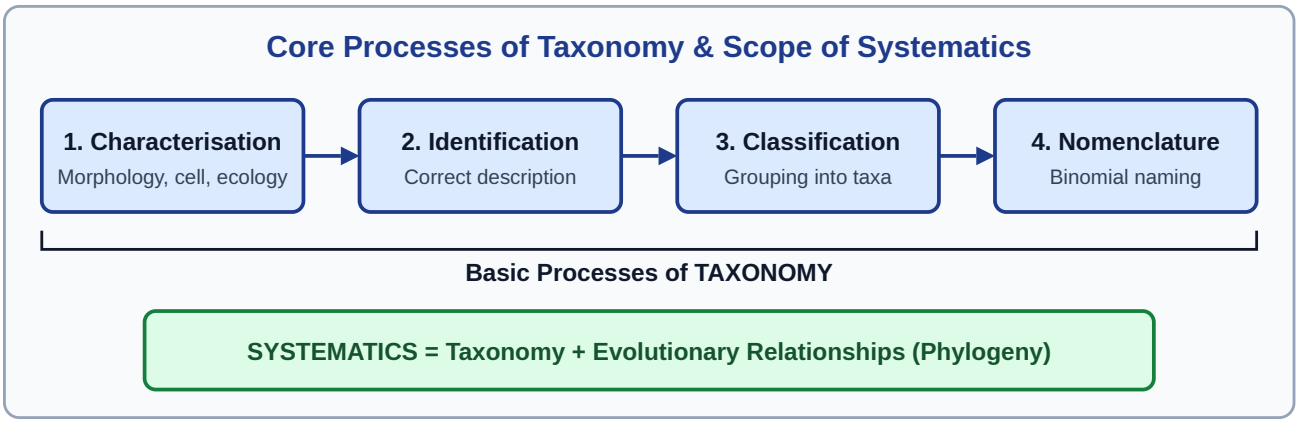
1. **Language and Font:** Biological names are generally in **Latin** and written in *italics*. They are Latinised or derived from Latin irrespective of their origin.
2. **Two Components:** The first word in a biological name represents the **genus**, while the second component denotes the **specific epithet**.
3. **Formatting (Handwritten vs. Printed):** Both the words in a biological name, when handwritten, are **separately underlined** (e.g., Mangifera indica), or printed in *italics* (e.g., *Mangifera indica*) to indicate their Latin origin.
4. **Capitalization:** The first word denoting the genus starts with a **capital letter**, while the specific epithet starts with a **small letter** (e.g., *Mangifera indica*, never *Mangifera Indica*).
5. **Author Citation:** The name of the author appears after the specific epithet (at the end of the biological name) and is written in an abbreviated form in Roman script (not italicized). For example: *Mangifera indica* Linn. indicates that this species was first described by Linnaeus.

3.4 Classification, Taxa, Taxonomy, and Systematics

- **Classification:** Since it is nearly impossible to study all living organisms individually, classification is the process by which anything is grouped into convenient categories based on some easily observable characters.
- **Taxa (singular: Taxon):** The scientific term for these convenient categories used to study organisms. Taxa can indicate categories at very different levels. For example, 'Plants', 'Wheat', 'Animals', 'Mammals', 'Dogs', 'Cats', 'Insects', and 'Alsatians' are all taxa. Because a dog is a mammal and mammals are animals, 'animals', 'mammals', and 'dogs' represent taxa at *different hierarchical levels*.
- **Taxonomy:** Based on characteristics, all living organisms can be classified into different taxa. This process of classification is called taxonomy.
  - **Basis of Earliest Classifications:** In early days, human beings needed to find sources for their basic needs of *food, clothing, and shelter*. Hence, the earliest classifications were based on the **'uses'** of various organisms.
  - **Basis of Modern Taxonomic Studies:** External and internal structure, along with the structure of the cell, development process, and ecological information of organisms are essential and form the basis of modern taxonomic studies.
  - **Four Basic Processes of Taxonomy:** *Characterisation, identification, classification, and nomenclature* are the processes that are basic to taxonomy.

- **Systematics:** Human beings have long been interested not only in knowing about different kinds of organisms and their diversities, but also the **relationships among them**. This branch of study is referred to as systematics.
  - Derived from the Latin word *'systema'*, which means systematic arrangement of organisms.
  - Carolus Linnaeus used *Systema Naturae* as the title of his publication.
  - The scope of systematics was later enlarged to include identification, nomenclature, and classification. Critically, **systematics takes into account evolutionary relationships between organisms**.

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Tutorial Diagram 1: Relationship between the four foundational processes of taxonomy and the broader scope of systematics.

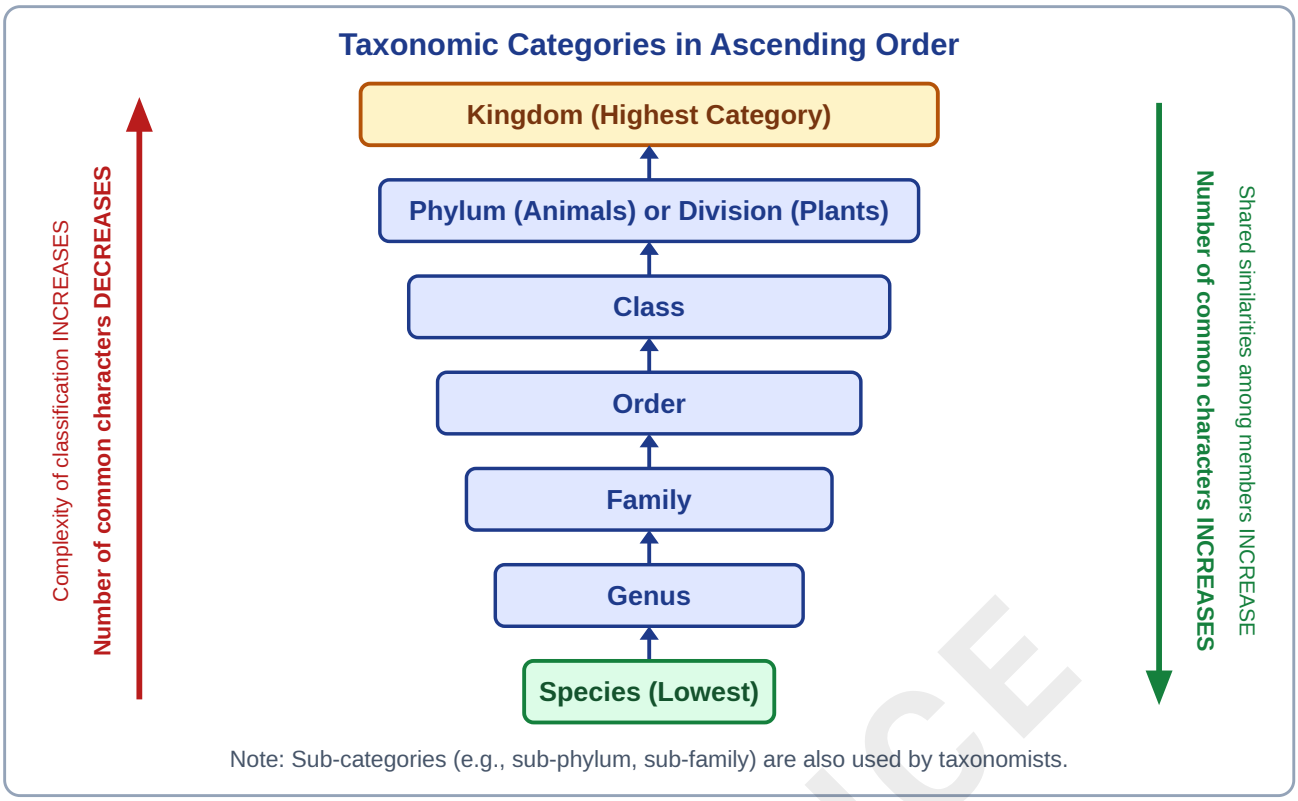
4. Taxonomic Categories and Hierarchy

Classification is not a single-step process; it involves a hierarchy of steps in which each step represents a **rank** or **category**.

- **Taxonomic Category:** Because each category is a part of the overall taxonomic arrangement, it is called a taxonomic category.
- **Taxonomic Hierarchy:** All taxonomic categories together constitute the taxonomic hierarchy.
- **Taxon (pl. taxa):** Each category, referred to as a unit of classification, represents a rank and is commonly termed a taxon. For example, insects represent a group of organisms sharing common features like **three pairs of jointed legs**. Thus, insects are recognizable concrete objects that can be classified and given a rank or category.

Crucial Conceptual Point: Taxonomic groups or categories are **distinct biological entities** and *not merely morphological aggregates*.

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Tutorial Diagram 2: Hierarchical arrangement of the seven broad taxonomic categories in ascending order and their character-sharing gradients.

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4.1 Detailed Breakdown of the Seven Obligate Categories

All organisms, including those in the plant and animal kingdoms, have **species as the lowest category**. Placing an organism into various categories requires knowledge of the characters of an individual or group of organisms so that similarities and dissimilarities can be identified.

1. Species

Taxonomic studies consider a group of individual organisms with **fundamental similarities** as a **species**. One can distinguish one species from another closely related species based on distinct morphological differences.

- In *Mangifera indica* (mango), *Solanum tuberosum* (potato), and *Panthera leo* (lion), the words *indica*, *tuberosum*, and *leo* represent the **specific epithets**, whereas *Mangifera*, *Solanum*, and *Panthera* are **genera**.
- Each genus may have one or more than one specific epithet representing different organisms that share morphological similarities:
  - Genus *Panthera* includes *leo* (lion), *tigris* (tiger), and *pardus* (leopard).
  - Genus *Solanum* includes *tuberosum* (potato), *nigrum* (black nightshade / makoi), and *melongena* (brinjal / eggplant).
  - Genus *Homo* includes the species *sapiens* (human beings: *Homo sapiens*).

2. Genus

Genus comprises a group of related species which has more characters in common in comparison to species of other genera. In other words, *genera are aggregates of closely related species*.

- **Plant Example:** Potato (*Solanum tuberosum*) and brinjal (*Solanum melongena*) are two different species, but both belong to the genus *Solanum*.
- **Animal Example:** Lion (*Panthera leo*), leopard (*Panthera pardus*), and tiger (*Panthera tigris*) share several common features and are all species of the genus *Panthera*. This genus differs from another genus, *Felis*, which includes domestic cats.

3. Family

Family has a group of related genera with still fewer similarities as compared to genus and species.

- **Basis in Plants:** Families are characterized on the basis of **both vegetative and reproductive features** of plant species. Three different genera—*Solanum*, *Petunia*, and *Datura*—are placed in the family **Solanaceae**.
- **Animal Example:** Genus *Panthera* (lion, tiger, leopard) is placed along with genus *Felis* (cats) in the family **Felidae**. By contrast, cats and dogs show some similarities and some differences; therefore, they are separated into two different families—**Felidae** (cats) and **Canidae** (dogs), respectively.

4. Order

Categories like species, genus, and families are based on a number of similar characters. Generally, **order and other higher taxonomic categories are identified based on the aggregates of characters**. An **Order** is an assemblage of families which exhibit a few similar characters (fewer in number as compared to different genera included in a family).

- **Plant Example:** Plant families like **Convolvulaceae** and **Solanaceae** are included in the order **Polymoniales** mainly based on their **floral characters**.
- **Animal Example:** The animal order **Carnivora** includes families like **Felidae** and **Canidae**.

5. Class

This category includes related orders.

- **Animal Example:** Order **Primata** (comprising monkey, gorilla, and gibbon) is placed in class **Mammalia** along with order **Carnivora** (which includes animals like tiger, cat, and dog). Class Mammalia also contains other orders.

6. Phylum / Division

Classes with common features constitute the next higher category.

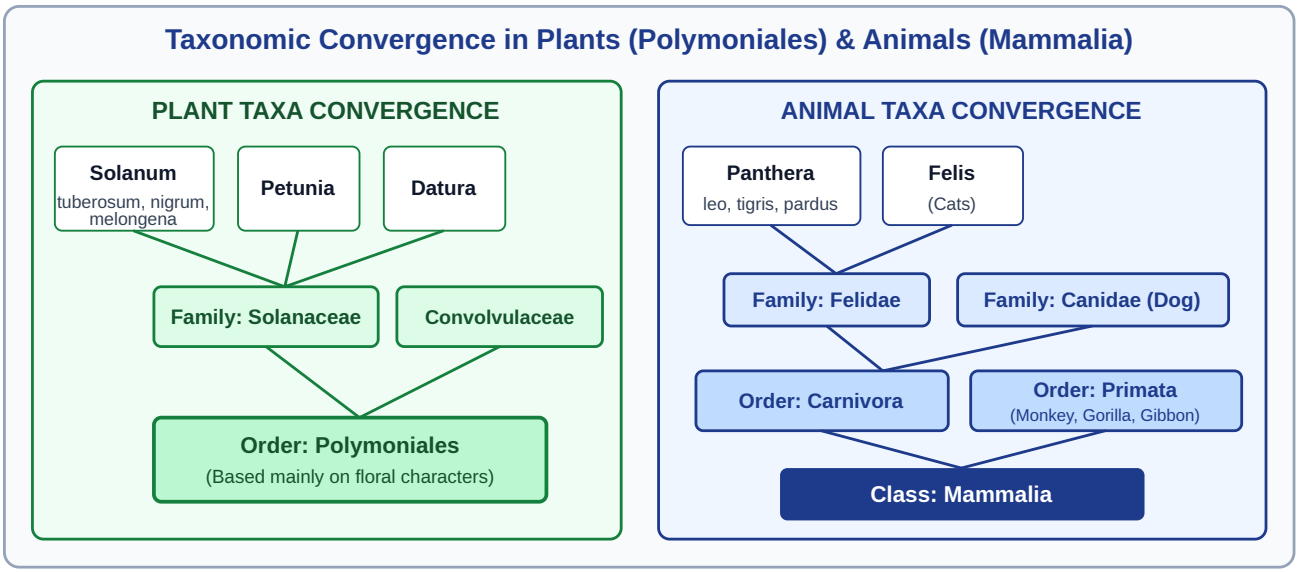
- **Phylum (in Animals):** Classes comprising animals like fishes, amphibians, reptiles, birds, along with mammals constitute a **Phylum**. All these, based on common features like the **presence of a notochord and a dorsal hollow neural system**, are included in phylum **Chordata**.
- **Division (in Plants):** In the case of plants, classes with a few similar characters are assigned to a higher category called **Division**.

7. Kingdom

The highest taxonomic category.

- **Kingdom Animalia:** All animals belonging to various phyla are assigned to Kingdom Animalia.
- **Kingdom Plantae:** Distinct from Animalia, it comprises all plants from various divisions.

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Tutorial Diagram 3: Grouping of related species into genera, families, orders, and classes in plants and animals.

4.2 Directional Trends Across the Hierarchy

- **Moving UPWARD from Species to Kingdom:** The number of common characteristics goes on **decreasing**. Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level. Hence, the problem of classification becomes more complex.
- **Moving DOWNWARD from Kingdom to Species:** Lower the taxa, **more are the characteristics** that the members within the taxon share.

4.3 Reference Table: Organisms with Their Taxonomic Categories

This table is one of the highest-yield factsheets for NEET UG questions. Memorize every cell and notice the standard suffixes for plant families (*-aceae*) and plant orders (*-ales*).

Common Name	Biological Name	Genus	Family	Order	Class	Phylum / Division
Man	*Homo sapiens*	*Homo*	Hominidae	Primata	Mammalia	Chordata (Phylum)
Housefly	*Musca domestica*	*Musca*	Muscidae	Diptera	Insecta	Arthropoda (Phylum)
Mango	*Mangifera indica*	*Mangifera*	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae (Division)
Wheat	*Triticum aestivum*	*Triticum*	Poaceae	Poales	Monocotyledonae	Angiospermae (Division)

5. Conceptual Connections & Worked Examples

Below are thorough analyses of foundational conceptual questions that build deeper clarity for competitive examinations.

Worked Example 1: Why are living organisms classified, and why do classification systems change periodically?

Analysis:

- **Why We Classify:** Earth is home to $1.7–1.8$ million known species. Studying every individual organism separately is impossible. Classification groups organisms into convenient categories (taxa)

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based on easily observable characters so that studying a representative member provides information about the entire group.

- **Why Systems Change:** Earliest classifications were utilitarian, based solely on the 'uses' of organisms for food, clothing, and shelter. Later, systems relied on gross morphological characters. As scientific knowledge expanded to include internal anatomy, cellular structure, developmental processes, ecological information, and evolutionary (phylogenetic) relationships, classification systems evolved to reflect true biological and evolutionary relationships. Moreover, new species are continuously being discovered as new and old areas are explored.

Worked Example 2: Evaluating Scientific Names and Taxonomic Sequences

Problem Statement:

(A) Between *Mangifera Indica* and *Mangifera indica*, identify the correctly written scientific name of mango and justify.

(B) Among the following sequences, identify the valid ascending sequence of taxonomic categories:

(i) Species → Order → Phylum → Kingdom

(ii) Genus → Species → Order → Kingdom

(iii) Species → Genus → Order → Phylum

Solution:

- **Part (A):** *Mangifera indica* is correct. According to the universal rules of binomial nomenclature, the first word denoting the genus (*Mangifera*) starts with a capital letter, while the second word denoting the specific epithet (*indica*) must start with a small letter.
- **Part (B):** Both **(i) Species → Order → Phylum → Kingdom** and **(iii) Species → Genus → Order → Phylum** follow the correct ascending directional order (even when intermediate ranks like Genus, Family, or Class are omitted). Sequence (ii) is incorrect because Species is lower than Genus and must precede Genus in an ascending sequence.

Worked Example 3: Meaning of 'Species' in Higher Plants/Animals vs. Bacteria

Analysis:

- **Taxonomic / Morphological Meaning:** A group of individual organisms with fundamental morphological similarities that can be distinguished from other closely related species based on distinct morphological differences.
- **Biological Species Concept (Ernst Mayr):** In sexually reproducing higher plants and animals, a species is a group of actually or potentially interbreeding natural populations that can produce fertile offspring and are reproductively isolated from other such groups.
- **Limitation in Bacteria:** Bacteria reproduce primarily asexually (by binary fission) and exhibit horizontal gene transfer. Therefore, the criterion of sexual interbreeding cannot be applied to define bacterial species; instead, bacterial taxonomy relies on cellular, biochemical, physiological, ecological, and genetic similarities.

6. NEET UG Practice Questions with Worked Solutions

The following original practice questions are strictly aligned with the NEET UG examination pattern—including single-choice MCQs, multi-statement evaluations, assertion-reason questions, and matrix-match items.

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Practice Question 1 (Single-Choice MCQ)

Who among the following evolutionary biologists was called "The Darwin of the 20th century" and was awarded the Triple Crown of Biology (Balzan Prize, International Prize for Biology, and Crafoord Prize)?

(1) Carolus Linnaeus

(2) Ernst Mayr

(3) R.H. Whittaker

(4) Alexander Agassiz

Correct Answer: (2)

Worked Solution: Ernst Mayr (1904–2004), the Harvard University evolutionary biologist, is known as "The Darwin of the 20th century". He pioneered the currently accepted definition of a biological species and received the Triple Crown of Biology: the Balzan Prize (1983), the International Prize for Biology (1994), and the Crafoord Prize (1999). Carolus Linnaeus established binomial nomenclature, and Alexander Agassiz is the name of the endowed professorship title Mayr held upon retirement.

Practice Question 2 (Multi-Statement Evaluation)

Read the following statements regarding the universal rules of binomial nomenclature:

I. Biological names are generally in Latin and written in italics; they are Latinised or derived from Latin irrespective of their origin.

II. When a biological name is handwritten, both the generic name and the specific epithet must be joined and underlined together with a single continuous line.

III. The first word denoting the genus starts with a capital letter, while the specific epithet starts with a small letter.

IV. The abbreviated name of the author appears before the generic name in italics.

Which of the above statements are **correct**?

(1) I and III only

(2) I, II, and III only

(3) II and IV only

(4) I, III, and IV only

Correct Answer: (1)

Worked Solution: Statements I and III are completely accurate rules of binomial nomenclature. Statement II is incorrect because when handwritten, both words in a biological name must be **separately underlined** (e.g., Mangifera indica). Statement IV is incorrect because the abbreviated name of the author appears **after the specific epithet** (at the end of the biological name) and is written in regular Roman script, e.g., *Mangifera indica* Linn.

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Practice Question 3 (Single-Choice MCQ)

Identify the correct expansion of the international code used by botanists to assign scientific names to plants:

(1) International Code of Biological Nomenclature

(2) International Code for Botanical Nomenclature

(3) International Congress of Botanical Naming

(4) International Committee for Biological Nomenclature

Correct Answer: (2)

Worked Solution: Scientific names for plants are based on agreed principles and criteria provided in the **International Code for Botanical Nomenclature (ICBN)**. For animals, taxonomists have evolved the **International Code of Zoological Nomenclature (ICZN)**.

Practice Question 4 (Assertion–Reason)

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The problem of classification becomes more complex as we move higher up from species to kingdom in the taxonomic hierarchy.

Reason (R): Higher the taxonomic category, fewer are the common characteristics shared by the members within the taxon, making it more difficult to determine relationships to other taxa at the same level.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (1)

Worked Solution: As we ascend from species to kingdom, the number of common characteristics goes on decreasing. Because members of a higher category share very few common features, determining the relationship of one taxon to other taxa at the same level becomes increasingly difficult, making classification more complex. Thus, both (A) and (R) are true and (R) directly explains (A).

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Practice Question 5 (Single-Choice MCQ)

Which of the following combinations of plant taxa are placed together in the family Solanaceae, and on what basis are plant families characterized?

(1) *Solanum*, *Petunia*, and *Datura*; based on both vegetative and reproductive features

(2) *Solanum*, *Mangifera*, and *Triticum*; based on vegetative features only

(3) *Petunia*, *Datura*, and *Convolvulaceae*; based on cell structure only

(4) *Solanum tuberosum*, *Mangifera indica*, and *Triticum aestivum*; based on floral characters only

Correct Answer: (1)

Worked Solution: In plants, families are characterized on the basis of **both vegetative and reproductive features** of plant species. The three different genera *Solanum*, *Petunia*, and *Datura* are placed in the family Solanaceae. (Note: Families Solanaceae and Convolvulaceae are placed in the order Polymoniales mainly based on floral characters.)

Practice Question 6 (Matrix-Match Question)

Match **Column I** (Taxonomic Category of Mango or Housefly) with **Column II** (Correct Taxon Name) and select the correct option:

Column I	Column II
A. Family of Mango	i. Diptera
B. Order of Mango	ii. Muscidae
C. Family of Housefly	iii. Sapindales
D. Order of Housefly	iv. Anacardiaceae

(1) A–iii, B–iv, C–i, D–ii

(2) A–iv, B–iii, C–ii, D–i

(3) A–iv, B–ii, C–iii, D–i

(4) A–ii, B–iii, C–iv, D–i

Correct Answer: (2)

Worked Solution: Mango (*Mangifera indica*) belongs to the Family **Anacardiaceae** (A–iv) and Order **Sapindales** (B–iii). Housefly (*Musca domestica*) belongs to the Family **Muscidae** (C–ii) and Order **Diptera** (D–i).

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Practice Question 7 (Single-Choice MCQ)

Consider the following statements about 'Dogs', 'Cats', 'Mammals', 'Wheat', and 'Animals':

(1) They are all taxa at the exact same hierarchical level.

(2) 'Animals', 'Mammals', and 'Dogs' represent taxa at different hierarchical levels.

(3) 'Wheat' is a category, whereas 'Mammals' is a morphological aggregate and not a taxon.

(4) 'Dogs' and 'Cats' belong to the same family Felidae.

Correct Answer: (2)

Worked Solution: 'Dogs', 'Cats', 'Mammals', 'Wheat', 'Rice', 'Plants', and 'Animals' are all convenient categories we use to study organisms, and the scientific term for these categories is **taxa**. However, taxa can indicate categories at very different levels: 'Dogs' (species/genus level), 'Mammals' (Class Mammalia), and 'Animals' (Kingdom Animalia) represent taxa at different levels. ('Dogs' belong to Canidae, while 'Cats' belong to Felidae.)

Practice Question 8 (Single-Choice MCQ)

What is the lowest taxonomic category shared in common by both **Mango** (*Mangifera indica*) and **Wheat** (*Triticum aestivum*)?

(1) Class — Dicotyledonae

(2) Order — Poales

(3) Division — Angiospermae

(4) Family — Poaceae

Correct Answer: (3)

Worked Solution: Let us compare the hierarchy of Mango and Wheat:
• **Mango:** Genus *Mangifera* → Family Anacardiaceae → Order Sapindales → Class Dicotyledonae → **Division Angiospermae**.
• **Wheat:** Genus *Triticum* → Family Poaceae → Order Poales → Class Monocotyledonae → **Division Angiospermae**.
They diverge at the Class level (Dicotyledonae vs. Monocotyledonae) and first converge at the **Division** level (**Angiospermae**).

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Practice Question 9 (Statement I & Statement II)

Given below are two statements:

Statement I: The word systematics is derived from the Greek word 'systema' and deals solely with the morphological description of a single species.

Statement II: The scope of systematics was enlarged to include identification, nomenclature, and classification, and it takes into account evolutionary relationships between organisms.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

Correct Answer: (4)

Worked Solution: Statement I is incorrect because the word *systematics* is derived from the **Latin** word *'systema'* (meaning systematic arrangement of organisms), not Greek, and it studies the diversity of organisms and the relationships among them. Statement II is completely correct.

Practice Question 10 (Single-Choice MCQ)

On the basis of which common features are classes comprising fishes, amphibians, reptiles, birds, and mammals included together in the Phylum Chordata?

(1) Presence of three pairs of jointed legs and chitinous exoskeleton

(2) Presence of external ears (pinnae) and body hair

(3) Presence of notochord and dorsal hollow neural system

(4) Presence of ventral solid neural system and pharyngeal gill slits

Correct Answer: (3)

Worked Solution: Classes comprising animals like fishes, amphibians, reptiles, birds, along with mammals are included in the next higher category called Phylum Chordata based on common features like the **presence of a notochord and a dorsal hollow neural system**. (External ears and body hair are features specific to Class Mammalia; three pairs of jointed legs characterize Class Insecta.)

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Practice Question 11 (Multi-Statement Evaluation)

Select the correct statements regarding the taxonomic positions of various animals:

A. Lion (*Panthera leo*), leopard (*P. pardus*), and tiger (*P. tigris*) belong to the same genus *Panthera*.

B. Genus *Panthera* and genus *Felis* are placed together in the family Canidae.

C. Order Primata (monkey, gorilla, gibbon) and Order Carnivora (tiger, cat, dog) are placed together in the class Mammalia.

D. Man (*Homo sapiens*) belongs to the family Hominidae and order Primata.

Choose the correct option:

(1) A, B, and C only

(2) A, C, and D only

(3) B and D only

(4) A, B, C, and D

Correct Answer: (2)

Worked Solution: Statements A, C, and D are correct. Statement B is false because genus *Panthera* and genus *Felis* (cats) are placed together in the family **Felidae**, whereas dogs are placed in a separate family, **Canidae**.

Practice Question 12 (Single-Choice MCQ)

Which of the following sets of features forms the essential basis of **modern taxonomic studies**?

(1) Only the economic uses of organisms for food, clothing, and shelter

(2) External and internal structure, structure of cell, development process, and ecological information of organisms

(3) Gross external morphology and local vernacular names only

(4) Deification of animate and inanimate objects evoking awe or fear

Correct Answer: (2)

Worked Solution: While the *earliest* classifications were based on the 'uses' of various organisms (for food, clothing, and shelter), **modern taxonomic studies** rely on external and internal structure, along with the structure of the cell, development process, and ecological information of organisms.

7. Common Exam Pitfalls & Revision Checklist

Common Pitfalls to Avoid in NEET UG

1. **ICBN vs. ICZN Prepositions:** Notice that ICBN is *International Code **for** Botanical Nomenclature*, whereas ICZN is *International Code **of** Zoological Nomenclature*.
2. **Underlining Rule:** When handwritten, the two words of a binomial name are **separately underlined**—never underlined with a single continuous line, and never italicized when handwritten (italics are for

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print).

3. **Author Citation Formatting:** In *Mangifera indica* Linn., "Linn." is abbreviated and printed in normal Roman script, *not* in italics.
4. **Phylum vs. Division:** The rank above Class is called **Phylum** in animals (e.g., Chordata, Arthropoda) and **Division** in plants (e.g., Angiospermae). Never interchange them in strict statement-based questions.
5. **Basis of Plant Family vs. Plant Order:** Plant **families** (like Solanaceae) are characterized on the basis of *both vegetative and reproductive features*, whereas plant **orders** (like Polymoniales) are identified mainly based on *floral characters* (aggregates of characters).
6. **Suffix Recognition in Plants:** In the reference table, do not confuse Family and Order names:
   - Plant **Families** end in **-aceae**: Solanaceae, Convolvulaceae, Anacardiaceae, Poaceae.
   - Plant **Orders** end in **-ales**: Polymoniales, Sapindales, Poales.

Rapid Revision Checklist

- ☑ Known and described biodiversity: $1.7–1.8$ million species.
- ☑ Ernst Mayr (1904–2004): "The Darwin of the 20th century"; Triple Crown: Balzan Prize (1983), International Prize for Biology (1994), Crafoord Prize (1999).
- ☑ Binomial nomenclature: Given by **Carolus Linnaeus**; publication titled *Systema Naturae*.
- ☑ Four basic processes of taxonomy: **Characterisation, Identification, Classification, Nomenclature**.
- ☑ Ascending hierarchy: **Species → Genus → Family → Order → Class → Phylum/Division → Kingdom**.
- ☑ *Solanum* species: *tuberosum* (potato), *nigrum* (makoi), *melongena* (brinjal).
- ☑ *Panthera* species: *leo* (lion), *tigris* (tiger), *pardus* (leopard).
- ☑ Family Solanaceae genera: *Solanum*, *Petunia*, *Datura*.
- ☑ Order Polymoniales families: Convolvulaceae + Solanaceae.
- ☑ Family Felidae genera: *Panthera* + *Felis*.
- ☑ Order Carnivora families: Felidae + Canidae.
- ☑ Class Mammalia orders: Primata + Carnivora (and others).

8. Concise Final Recap

The living world is rich in variety, with 1.7 to 1.8 million species identified and described, while many more remain unknown. To facilitate the study of this vast diversity—which is vital for agriculture, forestry, industry, and understanding our bio-resources—biologists have established universal rules for **identification, nomenclature, and classification** under international codes (ICBN and ICZN). Each organism is assigned a two-word Latinised scientific name (**Generic name** + **specific epithet**) under Carolus Linnaeus's **Binomial Nomenclature**. Organisms are grouped into distinct biological entities called **taxa** arranged in a seven-level **taxonomic hierarchy** from **Species** (lowest category, maximum shared characters) to **Kingdom** (highest category, minimum shared characters), while **systematics** integrates these taxonomic processes with the evolutionary relationships among organisms.

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